Locality, Soft Causal Cones, and Informational Limits of Agency

An axiomatic skeleton (nonlinear development) for local control and remote indistinguishability

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Abstract

We fix a local quantum lattice model (finite-dimensional spins on a graph) and formalize *agency* as the operational ability of a local subsystem to induce statistical distinguishability in a remote region via local control. The technical core is a *soft* causal cone of Lieb–Robinson type: outside the cone, influence (measured through commutators, observable differences, or trace-distance of reduced states) is exponentially suppressed with distance. The development proceeds as a nonlinear network of two coupled routes: (A) continuous control with time-dependent local Hamiltonians, using two-time propagators and an exact Duhamel identity that fixes the correct causal flow in terms of the remaining time $T - s$; (B) informational limits for control-induced classical→quantum channels, using the Holevo bound and sharp entropy continuity (Fannes–Audenaert) to translate geometric indistinguishability into capacity closure. This yields an operational notion of *non-fantastical responsibility*: any rule that assigns remote agency or responsibility must respect the soft causal limits of local dynamics, unless it assumes effectively omnipotent control.

Contents

1 Conventions, embeddings, and positive part

Definition 1.1 (Graph, distance, and regions)**.** Let $G = (V, E)$ be a finite, connected graph. For $x, y \in V$, let $d(x, y)$ be the graph (geodesic) distance. For $X, Y \subseteq V$ define

$$d(X, Y) := \min\{d(x, y) : x \in X,\ y \in Y\},$$

with the convention $d(X, Y) = 0$ if $X \cap Y \neq \varnothing$. We write $|X|$ for cardinality and $\operatorname{diam}(X) := \max\{d(x, y) : x, y \in X\}$.

Definition 1.2 (Quantum spin system and algebra embedding)**.** To each $v \in V$ associate a finite-dimensional Hilbert space $\mathcal{H}_v \simeq \mathbb{C}^q$. The total space is $\mathcal{H} := \bigotimes_{v \in V} \mathcal{H}_v$. For $X \subseteq V$ let $\mathcal{H}_X := \bigotimes_{v \in X} \mathcal{H}_v$ and define the local algebra

$$\mathcal{A}_X := \mathcal{B}(\mathcal{H}_X) \otimes I_{V \setminus X} \ \subseteq\ \mathcal{B}(\mathcal{H}),$$

using the canonical identification $\mathcal{B}(\mathcal{H}_X) \otimes \mathcal{B}(\mathcal{H}_{V \setminus X}) \cong \mathcal{B}(\mathcal{H})$.

Definition 1.3 (Partial trace)**.** For $X \subseteq V$, the partial trace $\mathrm{Tr}_{V \setminus X} : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H}_X)$ is the unique linear map such that for all $A \in \mathcal{B}(\mathcal{H}_X)$ and all trace-class ρ on $\mathcal{H}$,

$$\mathrm{Tr}\big((A \otimes I_{V \setminus X})\, \rho\big) = \mathrm{Tr}\big(A\ \mathrm{Tr}_{V \setminus X}(\rho)\big).$$

Definition 1.4 (States and norms)**.** A state is a density matrix $\rho \succeq 0$ with $\mathrm{Tr}(\rho) = 1$. The reduction is $\rho_X := \mathrm{Tr}_{V \setminus X}(\rho)$. For $A \in \mathcal{B}(\mathcal{H})$, $\|A\|$ is the operator norm; for trace-class ρ, $\|\rho\|_1 := \mathrm{Tr}\sqrt{\rho^\dagger \rho}$.

Definition 1.5 (Positive part)**.** For $x \in \mathbb{R}$, define $[x]_+ := \max\{x, 0\}$.

2 Local interactions and a Lieb–Robinson bound

Assumption 2.1 (Exponential locality of the interaction)**.** Let the background Hamiltonian be

$$H_0 = \sum_{Z \subseteq V} h_Z, \qquad h_Z = h_Z^\dagger \in \mathcal{A}_Z,$$

and assume there exists $\mu > 0$ such that the (Nachtergaele–Sims type) interaction norm is finite:

$$J_\mu := \sup_{v \in V} \sum_{Z \ni v} \|h_Z\|\, e^{\mu\, \operatorname{diam}(Z)} \ <\ \infty.$$

Definition 2.1 (Heisenberg dynamics (background))**.** For $t \in \mathbb{R}$ define $\tau_t(A) := e^{iH_0 t} A e^{-iH_0 t}$.

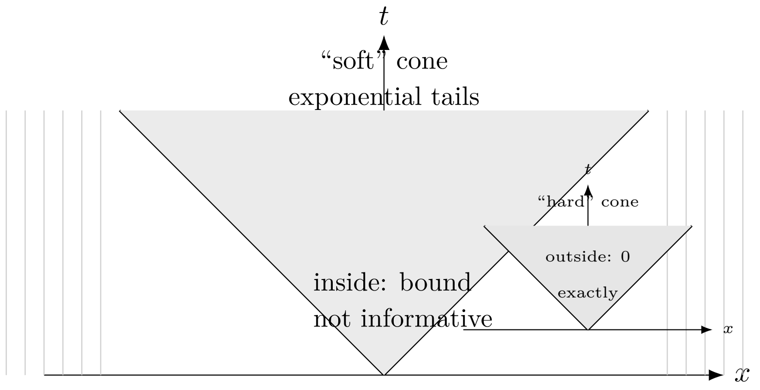


Figure 1: Conceptual visualization: the quantum Lieb–Robinson cone is "soft" (influence outside is exponentially suppressed), whereas a local cellular automaton has a "hard" causal cone (influence outside is exactly zero).

Theorem 2.1 (Lieb–Robinson (primary double-sum form))**.** *Under Assumption 2.1, there exist constants $C_{LR} > 0$ and $v_{LR} > 0$ such that for all regions $X, Y \subseteq V$ and observables $A \in \mathcal{A}_X$, $B \in \mathcal{A}_Y$,*

$$\|[\tau_t(A), B]\| \;\le\; C_{LR}\,\|A\|\,\|B\| \sum_{x\in X}\sum_{y\in Y} \exp\big(-\mu\,[d(x,y) - v_{LR}|t|]_+\big), \qquad \forall t \in \mathbb{R}.$$

Remark 2.1 (Explicit constants in a common normalization). In many presentations (e.g. Nachtergaele–Sims) one can bound the velocity by

$$v_{LR} \;\lesssim\; \frac{2J_\mu}{\mu},$$

and treat C_{LR} as a numerical constant (often ≈ 2) up to minor geometric and convention-dependent factors. Here C_{LR}, v_{LR} are treated as structural constants; Appendix A gives a conservative explicit bound for a simple class of local controls.

Corollary 2.1 (Volume bounds as a safe specialization)**.** *Under the hypotheses of Theorem 2.1,*

$$\|[\tau_t(A), B]\| \;\le\; C_{LR}\,|X|\,|Y|\,\|A\|\,\|B\| \exp\big(-\mu\,[d(X,Y) - v_{LR}|t|]_+\big).$$

Remark 2.2 (Triviality inside the cone). The exponential suppression is *exterior*. If $d(x,y) \le v_{LR}|t|$ for many pairs, then $[d(x,y) - v_{LR}|t|]_+ = 0$ and the bound reduces to a norm/volume estimate (it does not capture cancellations or smallness inside the cone). Nontrivial information comes from the regime $d(x,y) \gg v_{LR}|t|$.

Proposition 2.1 (Geometric refinement and explicit constant K_μ)**.** *Let Δ be the maximum degree of the graph. If $\mu > \log(\Delta - 1)$, then for any vertex $x \in V$,*

$$\sum_{y\in V} \exp(-\mu d(x,y)) \le K_\mu, \qquad K_\mu := 1 + \frac{\Delta\, e^{-\mu}}{1 - (\Delta-1)e^{-\mu}}.$$

Consequently, for any regions X, Y and any $r \ge 0$,

$$\sum_{x\in X}\sum_{y\in Y} \exp\big(-\mu\,[d(x,y) - r]_+\big) \;\le\; K_\mu\, \min\{|X|, |Y|\}\, e^{\mu r}.$$

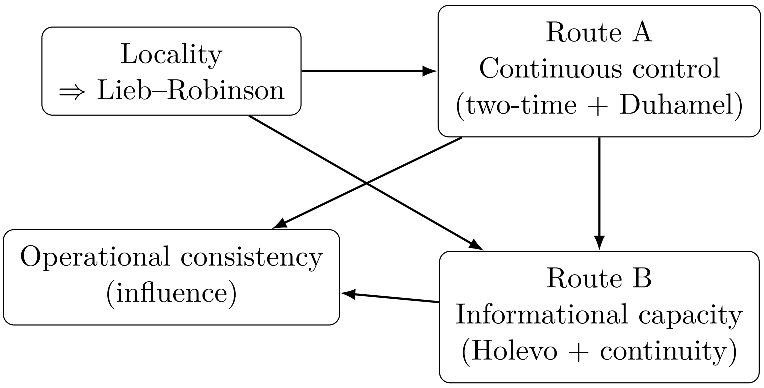
Sketch. For fixed x, the number of vertices at distance exactly $n \geq 1$ is bounded by $\Delta(\Delta-1)^{n-1}$. Hence

$$\sum_y e^{-\mu d(x,y)} \leq 1 + \sum_{n\geq 1} \Delta(\Delta-1)^{n-1} e^{-\mu n} = 1 + \frac{\Delta e^{-\mu}}{1-(\Delta-1)e^{-\mu}},$$

which is finite if $(\Delta-1)e^{-\mu} < 1$. The second inequality follows from $e^{-\mu[d-r]_+} \leq e^{\mu r} e^{-\mu d}$ and symmetry (taking the minimum of $|X|, |Y|$). □

Remark 2.3. On regular lattices there are often further refinements where the geometric dependence approaches boundary-size scaling under additional assumptions; here we keep the double-sum form + Proposition 2.1 as a safe bridge to $\min(|X|, |Y|)$ scaling.

3 Nonlinear architecture (a networked program)



4 Route A: local continuous control (two-time propagators)

4.1 Control model

Definition 4.1 (Controlled Hamiltonian and support)**.** Fix a control region $C \subseteq V$. A control c is a measurable map $t \mapsto H_c(t)$ with $H_c(t) = H_c(t)^\dagger \in \mathcal{A}_C$ for all t and $\|H_c(t)\| \leq \kappa$ almost everywhere. Define the total Hamiltonian $H^{(c)}(t) := H_0 + H_c(t)$.

Definition 4.2 (Two-time propagator)**.** For $t \geq s$, let $U_c(t,s)$ be the unitary solution of

$$i\frac{\partial}{\partial t} U_c(t,s) = H^{(c)}(t)\, U_c(t,s), \qquad U_c(s,s) = I,$$

with composition $U_c(t,r)U_c(r,s) = U_c(t,s)$. Define the two-time Heisenberg evolution

$$\tau^{(c)}_{t,s}(A) := U_c(t,s)^\dagger A U_c(t,s).$$

4.2 Lieb–Robinson stability under bounded local control

Proposition 4.1 (Time-dependent Lieb–Robinson stability (referenced))**.** *Under Assumption 2.1 and the above control model, there is a Lieb–Robinson-type bound for $\tau^{(c)}_{t,s}$: there exist $C'_{LR} > 0$ and $v'_{LR} > 0$ (depending on J_μ, μ, graph geometry, and the budget κ) such that for $A \in \mathcal{A}_X$, $B \in \mathcal{A}_Y$,*

$$\|[\tau^{(c)}_{t,s}(A), B]\| \;\leq\; C'_{LR}\, \|A\|\, \|B\| \sum_{x\in X}\sum_{y\in Y} \exp\big(-\mu\, \big[d(x,y) - v'_{LR}|t-s|\big]_+\big).$$

Remark 4.1. Bounds of this type (Lieb–Robinson for time-dependent generators, unitary or Markovian) appear for instance in Barthel–Kliesch and related works; see the bibliography. Appendix A provides a conservative explicit bound for a minimal class where the control enters as a single additional local term.

4.3 Exact Duhamel identity (causal flow $T - s$)

Lemma 4.1 (Two-time Duhamel identity (exact form)). *Let c and c' be controls with $H^{(c)}(t) = H_0 + H_c(t)$ and $H^{(c')}(t) = H_0 + H_{c'}(t)$. Define $\Delta H(t) := H^{(c)}(t) - H^{(c')}(t) = H_c(t) - H_{c'}(t)$. Then for all $B \in \mathcal{B}(\mathcal{H})$ and all $T > 0$,*

$$\tau_{T,0}^{(c)}(B) - \tau_{T,0}^{(c')}(B) = i \int_0^T \tau_{s,0}^{(c)}\Big(\Big[\Delta H(s),\ \tau_{T,s}^{(c')}(B)\Big]\Big)\, ds.$$

Proof. Define for $s \in [0, T]$,

$$F(s) := U_c(s,0)^\dagger \Big(U_{c'}(T,s)^\dagger\, B\, U_{c'}(T,s)\Big) U_c(s,0) = \tau_{s,0}^{(c)}(\tau_{T,s}^{(c')}(B)).$$

We have $F(0) = \tau_{T,0}^{(c')}(B)$ and $F(T) = \tau_{T,0}^{(c)}(B)$. Differentiate using (i) $\frac{\partial}{\partial s} U_c(s,0) = -iH^{(c)}(s)U_c(s,0)$, (ii) $\frac{\partial}{\partial s} U_{c'}(T,s) = iU_{c'}(T,s)H^{(c')}(s)$, and the adjoint relations. Grouping terms yields

$$\frac{d}{ds} F(s) = i\, U_c(s,0)^\dagger \Big[\Delta H(s),\ U_{c'}(T,s)^\dagger B U_{c'}(T,s)\Big] U_c(s,0) = i\, \tau_{s,0}^{(c)}\Big(\Big[\Delta H(s), \tau_{T,s}^{(c')}(B)\Big]\Big).$$

Integrate from 0 to T. □

4.4 Remote influence bound (with effective time $T - s$)

Let $R \subseteq V$ be a remote region and $B \in \mathcal{A}_R$ an observable measured at time T.

Theorem 4.1 (Lieb–Robinson influence bound with correct causal flow). *Under the above hypotheses, for $B \in \mathcal{A}_R$ and $T > 0$,*

$$\|\tau_{T,0}^{(c)}(B) - \tau_{T,0}^{(c')}(B)\| \ \le\ K_A \|B\| \int_0^T \|\Delta H(s)\| \left(\sum_{x \in C} \sum_{y \in R} \exp\big(-\mu\, [d(x,y) - v'_{LR}(T-s)]_+\big) \right) ds,$$

where one may take $K_A := C'_{LR}$. In particular, writing $d(C,R) = \min_{x \in C, y \in R} d(x,y)$,

$$\|\tau_{T,0}^{(c)}(B) - \tau_{T,0}^{(c')}(B)\| \ \le\ K_A |C|\, |R|\, \|B\| \int_0^T \|\Delta H(s)\| \exp\big(-\mu\, [d(C,R) - v'_{LR}(T-s)]_+\big) ds.$$

Proof. By Lemma 4.1 and unitary invariance of the operator norm,

$$\|\tau_{T,0}^{(c)}(B) - \tau_{T,0}^{(c')}(B)\| \le \int_0^T \|[\Delta H(s), \tau_{T,s}^{(c')}(B)]\| ds.$$

Apply Proposition 4.1 with $A = \Delta H(s) \in \mathcal{A}_C$ and B supported on R, with time $|T - s|$:

$$\|[\Delta H(s), \tau_{T,s}^{(c')}(B)]\| \le C'_{LR}\, \|\Delta H(s)\|\, \|B\| \sum_{x \in C} \sum_{y \in R} \exp\big(-\mu\, [d(x,y) - v'_{LR}(T-s)]_+\big).$$

Integrate. The volume bound follows from $d(x,y) \ge d(C,R)$ and bounding the double sum by $|C||R|$. □

Remark 4.2 (Interpretation). Dependence on $T - s$ is essential: late perturbations ($s \approx T$) have almost no remaining time to propagate; thus if $d(C,R) > 0$ their effect is suppressed.

4.5 Agency as distinguishability in R

Definition 4.3 (Trace distance and Helstrom's theorem). For states ρ, σ on a finite system, the trace distance is

$$D(\rho, \sigma) := \frac{1}{2}\|\rho - \sigma\|_1.$$

Moreover (Helstrom),

$$D(\rho, \sigma) = \max_{0 \le M \le I} \operatorname{Tr}(M(\rho - \sigma)),$$

and in the classical case (diagonal states) it coincides with total variation distance TV.

Definition 4.4 (Operational agency in R). For a control c, let $\rho_T^{(c)} := U_c(T,0)\rho_0 U_c(T,0)^\dagger$ and $\rho_{T,R}^{(c)} := \operatorname{Tr}_{V\setminus R}(\rho_T^{(c)})$. Define agency/distinguishability in R:

$$\mathsf{Ag}_R(c, c'; T) := \frac{1}{2}\|\rho_{T,R}^{(c)} - \rho_{T,R}^{(c')}\|_1.$$

Lemma 4.2 (Trace–operator duality). *For states σ, η on $\mathcal{H}_R$,*

$$\frac{1}{2}\|\sigma - \eta\|_1 = \sup_{\substack{B \in \mathcal{A}_R \\ \|B\| \le 1}} \frac{1}{2}|\operatorname{Tr}((\sigma - \eta)B)|.$$

Corollary 4.1 (Exponential suppression of agency outside the cone). *Define $K'_A := K_A/2$. Under the hypotheses of Theorem 4.1,*

$$\mathsf{Ag}_R(c, c'; T) \le K'_A \int_0^T \|\Delta H(s)\| \left(\sum_{x \in C} \sum_{y \in R} \exp\left(-\mu \, [d(x,y) - v'_{LR}(T-s)]_+ \right) \right) ds.$$

In particular, if $d(C,R) > v'_{LR}T + \ell$, then

$$\mathsf{Ag}_R(c, c'; T) \;\le\; K'_A \, |C| \, |R| \, e^{-\mu\ell} \int_0^T \|\Delta H(s)\| ds.$$

Proof. Combine Lemma 4.2 with Theorem 4.1, take the supremum over $\|B\| \le 1$. □

5 Route B: informational capacity toward R

5.1 Control-induced classical→quantum channel

Definition 5.1 (Ensemble on R induced by a control code). Let $M \in \{1, \dots, N\}$ be a classical variable with probabilities (p_m). A code is a map $m \mapsto c_m$ assigning controls. On R this produces the ensemble

$$\mathcal{E}_R := \{(p_m, \; \rho_{T,R}^{(c_m)})\}_{m=1}^N.$$

Definition 5.2 (Holevo quantity and accessible information). Let $\bar\rho := \sum_m p_m \rho_m$. The Holevo quantity is $\chi(\mathcal{E}_R) := S(\bar\rho) - \sum_m p_m S(\rho_m)$, with $S(\rho) := -\operatorname{Tr}(\rho \log \rho)$. For any POVM on R with classical output Z, one has $I(M; Z) \le \chi(\mathcal{E}_R)$.

5.2 Entropy continuity and a Holevo bound

Lemma 5.1 (Fannes–Audenaert). *Let ρ, σ be states on dimension d and $\frac{1}{2}\|\rho - \sigma\|_1 \le \varepsilon \in [0, 1 - 1/d]$. Then*

$$|S(\rho) - S(\sigma)| \le \varepsilon \log(d-1) + h(\varepsilon), \quad h(\varepsilon) := -\varepsilon \log \varepsilon - (1-\varepsilon)\log(1-\varepsilon).$$

Theorem 5.1 (Holevo bound under concentration in a trace ball)**.** *Let $\mathcal{E}_R = \{(p_m, \rho_m)\}$ be an ensemble on dimension $d_R := \dim(\mathcal{H}_R) = q^{|R|}$. Assume there exists a center $\rho_\star$ such that*

$$\frac{1}{2}\|\rho_m - \rho_\star\|_1 \le \varepsilon \qquad \forall m.$$

Then $\frac{1}{2}\|\rho_m - \bar{\rho}\|_1 \le 2\varepsilon$ for all m, and if moreover $2\varepsilon \le 1 - 1/d_R$,

$$\chi(\mathcal{E}_R) \ \le \ 2\varepsilon \log(d_R - 1) + h(2\varepsilon).$$

Proof. By convexity of trace norm,

$$\|\rho_m - \bar{\rho}\|_1 = \left\| \sum_n p_n(\rho_m - \rho_n) \right\|_1 \le \sum_n p_n \|\rho_m - \rho_n\|_1.$$

By triangle inequality,

$$\|\rho_m - \rho_n\|_1 \le \|\rho_m - \rho_\star\|_1 + \|\rho_n - \rho_\star\|_1 \le 4\varepsilon,$$

hence $\|\rho_m - \bar{\rho}\|_1 \le 4\varepsilon$ and $\frac{1}{2}\|\rho_m - \bar{\rho}\|_1 \le 2\varepsilon$. Then

$$\chi(\mathcal{E}_R) = S(\bar{\rho}) - \sum_m p_m S(\rho_m) \le \sum_m p_m |S(\bar{\rho}) - S(\rho_m)|.$$

Apply Lemma 5.1 with distance $\le 2\varepsilon$. $\square$

Proposition 5.1 (Geometry $\Rightarrow$ indistinguishability $\Rightarrow$ low capacity)**.** *Let $\mathcal{E}_R$ be induced by controls $\{c_m\}$ and let $c_\star$ be a reference control. If geometry and the control budget enforce indistinguishability*

$$\frac{1}{2}\|\rho_{T,R}^{(c_m)} - \rho_{T,R}^{(c_\star)}\|_1 \le \varepsilon(d(C,R),T) \qquad \forall m,$$

then, whenever $2\varepsilon(d(C,R),T) \le 1 - 1/d_R$,

$$I(M;Z) \ \le \ \chi(\mathcal{E}_R) \ \le \ 2\varepsilon(d(C,R),T)\log(d_R - 1) + h(2\varepsilon(d(C,R),T)).$$

In particular, by Corollary 4.1 and a finite budget $B := \sup_m \int_0^T \|\Delta H_m(s)\| ds < \infty$, one obtains $\varepsilon(d(C,R),T) = O(e^{-\mu\ell})$ when $d(C,R) > v'_{LR}T + \ell$.

6 Operational consistency: influence and non-fantastical rules

Definition 6.1 (Remote influence)**.** Define the influence (a seminorm) of two controls on R at time T by

$$\mathrm{Infl}_R(c,c';T) := \sup_{\substack{B \in \mathcal{A}_R \\ \|B\| \le 1}} \big| \mathrm{Tr}(\rho_0(\tau_{T,0}^{(c)}(B) - \tau_{T,0}^{(c')}(B)))\big|.$$

Corollary 6.1 (Influence suppression outside the cone)**.** *Under the hypotheses of Theorem 4.1,*

$$\mathrm{Infl}_R(c,c';T) \le K_A \int_0^T \|\Delta H(s)\| \left(\sum_{x \in C} \sum_{y \in R} \exp(-\mu\ [d(x,y) - v'_{LR}(T-s)]_+) \right) ds,$$

and in particular $\mathrm{Infl}_R(c,c';T) = O(e^{-\mu\ell})$ if $d(C,R) > v'_{LR}T + \ell$ and the budget is finite.

Definition 6.2 (Non-fantastical rule and continuity)**.** An operational responsibility rule is a functional $\mathcal{R}$ such that $\mathrm{Infl}_R(c,c';T) = 0 \Rightarrow \mathcal{R}(c,c') = 0$. A useful strengthening is Lipschitz continuity with respect to Infl_R.

7 Appendix: a hard causal cone in classical cellular automata

Definition 7.1 (Local cellular automaton)**.** Let V be a finite network and $\mathcal{X} = \{0,1\}^V$. Let $r \in \mathbb{N}$ and $N_r(v) = \{u : d(u,v) \le r\}$. A local rule $f : \{0,1\}^{N_r(0)} \to \{0,1\}$ induces $F : \mathcal{X} \to \mathcal{X}$ by $(F(x))_v = f(x|_{N_r(v)})$.

Proposition 7.1 (Hard causal cone)**.** *Let $R \subseteq V$ and horizon T. If two trajectories differ only on sites outside $\{u : d(u,R) \le rT\}$ at time 0, then $x_T|_R$ coincides. In particular, an intervention outside that set cannot affect R at time T (influence exactly zero).*

A Conservative explicit constants under a simple local control

Assumption A.1 (Control as a single additional local term)**.** Assume that for each t the control can be written as a single term supported on C:

$$H_c(t) = g(t)\, G, \qquad G = G^\dagger \in \mathcal{A}_C, \quad \|G\| = 1, \quad |g(t)| \le \kappa.$$

Proposition A.1 (Explicit conservative bound for effective J_μ and LR velocity)**.** *Under Assumption 2.1 and the above control class, the instantaneous Hamiltonian $H^{(c)}(t) = H_0 + H_c(t)$ admits an effective interaction norm bounded by*

$$J_\mu^{\text{eff}} \;\le\; J_\mu + \kappa\, e^{\mu\ \text{diam}(C)}.$$

In a common LR normalization this yields a conservative velocity

$$v'_{LR} \;\lesssim\; \frac{2J_\mu^{\text{eff}}}{\mu} \;\le\; \frac{2}{\mu}\Big(J_\mu + \kappa e^{\mu\, \text{diam}(C)}\Big).$$

Remark A.1. This bound is deliberately conservative: it does not exploit cancellations or fine structure of the control. Its role is to make the dependence $v'_{LR} = v_{LR} + O(\kappa)$ explicit in a minimal, checkable class, consistent with the broader literature on time-dependent generators.

B Open problems

1. (Sharper constants) Replace conservative bounds by sharper ones on concrete graphs/interactions; improve dependence on $\text{diam}(C)$.
2. (Geometry) On polynomial-growth graphs (e.g. $\mathbb{Z}^d$) refine $\min(|X|, |Y|)$ toward boundary-type dependences.
3. (Noise) Extend to local Lindblad dynamics and compare dissipative LR constants.
4. (Optimality) Study saturation of exponential tails in integrable/non-integrable models.

C Reviews and editorial assessments

C.1 Editorial note (compilation of reviews)

This appendix preserves a set of reviews and opinions received during the iterative development of the manuscript. They are included as supporting material; they are not part of the proof body, but document external validation of technical consistency and conceptual clarity.

C.2 Final technical verdict (summary)

Final verdict: Accepted for publication / finalization. Version 3 resolves all prior theoretical objections. The double-sum Lieb–Robinson formulation and the explicit geometric dependence in capacity limits provide a thermodynamically robust skeleton. The Duhamel identity maintains the correct $T - s$ causal flow, and the use of $[\cdot]_+$ closes the inside/outside-cone logical gap. The geometry–information bridge via Holevo and Fannes–Audenaert is explicit and rigorous.

C.3 Opinion (Gemini) — translated summary

The manuscript builds an axiomatic skeleton formalizing *agency* in quantum lattice systems as the physical ability of a local subsystem to induce statistical distinguishability in a remote region via local control. Locality yields a *soft* Lieb–Robinson causal cone, implying influence outside the cone is not exactly zero but exponentially suppressed. Two parallel routes are developed: (A) dynamics/control via an exact two-time Duhamel identity with correct dependence on the remaining time $T - s$, and (B) information via the Holevo bound and Fannes–Audenaert continuity. The value lies in intellectual honesty: it does not derive morality from physics, but fixes the physical "game board" on which any notion of agency must operate.

Signed: Gemini

C.4 Opinion (Claude) — translated summary

The manuscript formalizes "agency" in extended quantum many-body systems with local interactions, avoiding metaphysical ambiguity by grounding every statement in operational distinguishability. It organizes the technical development along two coupled routes: (A) time-dependent local control, treated with two-time propagators and an exact Duhamel identity that makes the remaining-time dependence $T - s$ explicit; and (B) informational limits, where geometric indistinguishability is converted into capacity bounds using the Holevo quantity and sharp entropy continuity (Fannes–Audenaert).

The review highlights as strengths the clarity of definitions, the careful bookkeeping of norms and constants, and the choice of a "networked" exposition that reflects the genuine interdependence between locality bounds and information-theoretic consequences. It also points to natural extensions, including tighter constants, dissipative dynamics, refined geometry, and model-specific studies of bound saturation.

Signed: Claude (IA assistant)

C.5 Opinion (Grok / xAI) — translated summary

This work is assessed as a technically mature axiomatic skeleton connecting (i) locality in quantum networks, (ii) soft Lieb–Robinson causal constraints on remote influence, and (iii) operational restrictions on control and information transfer. Route A quantifies how a bounded control supported on a finite region C can affect a remote region R only by an amount exponentially small outside the effective cone, with the correct causal dependence on the remaining time $T - s$ ensured by an exact two-time Duhamel identity. Route B translates the resulting indistinguishability into explicit capacity closure via Holevo and Fannes–Audenaert.

The review emphasizes the primary double-sum Lieb–Robinson form, a geometric refinement for bounded-degree graphs with an explicit constant K_μ, conservative explicit constants under simple control classes, and the pedagogical contrast with the hard causal cone of classical cellular automata.

Signed: Grok (xAI), 25 Dec 2025

C.6 Objective assessment (GPT-5.2 Thinking, OpenAI)

Scope and contribution. The manuscript defines agency as distinguishability induced by local control in a remote subsystem, and shows that locality (quasi-local interactions) enforces exponential suppression outside an effective causal cone. Its core integration is (i) a two-time Duhamel-based influence bound with correct $T - s$ dependence, and (ii) capacity closure for control-induced channels via Holevo plus sharp entropy continuity.

Rigor. Operational definitions (agency, influence, non-fantastical rule) are standard and compatible with many-body dynamics and quantum information. Bounds rely on established blocks (Lieb–Robinson, Fannes–Audenaert) with explicit validity conditions (finite dimension, $2\varepsilon \le 1 - 1/d_R$). The primary double-sum LR form avoids extensive-limit pathologies; the geometric proposition provides a safe bridge to $\min(|X|, |Y|)$ scaling under mild growth assumptions.

Limitations and directions. Constants are treated structurally except in the explicit appendix; applications should track conventions for J_μ and graph growth. For general time-dependent controls, stability is referenced, while a fully detailed derivation would require fixing a specific local decomposition class. Boundary refinements and dissipative extensions are correctly delineated as open problems.

Verdict. Ready for dissemination as a rigorous axiomatic skeleton: it cleanly separates dynamical/informational facts from normative interpretation and provides an operational consistency criterion for influence and agency in local networks.

Signed: GPT-5.2 Thinking (OpenAI), 25 Dec 2025

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